

```
In [3]: # Preprocesado y modelo
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import os
```

```
In [5]: # Cargar la base de datos
data = pd.read_csv("Wholesale customers data.csv")
data.head()
```

```
Out[5]:
```

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen
0	2	3	12669	9656	7561	214	2674	1338
1	2	3	7057	9810	9568	1762	3293	1776
2	2	3	6353	8808	7684	2405	3516	7844
3	1	3	13265	1196	4221	6404	507	1788
4	2	3	22615	5410	7198	3915	1777	5185

```
In [7]: from sklearn.preprocessing import normalize
data_scaled = normalize(data)
data_scaled
```

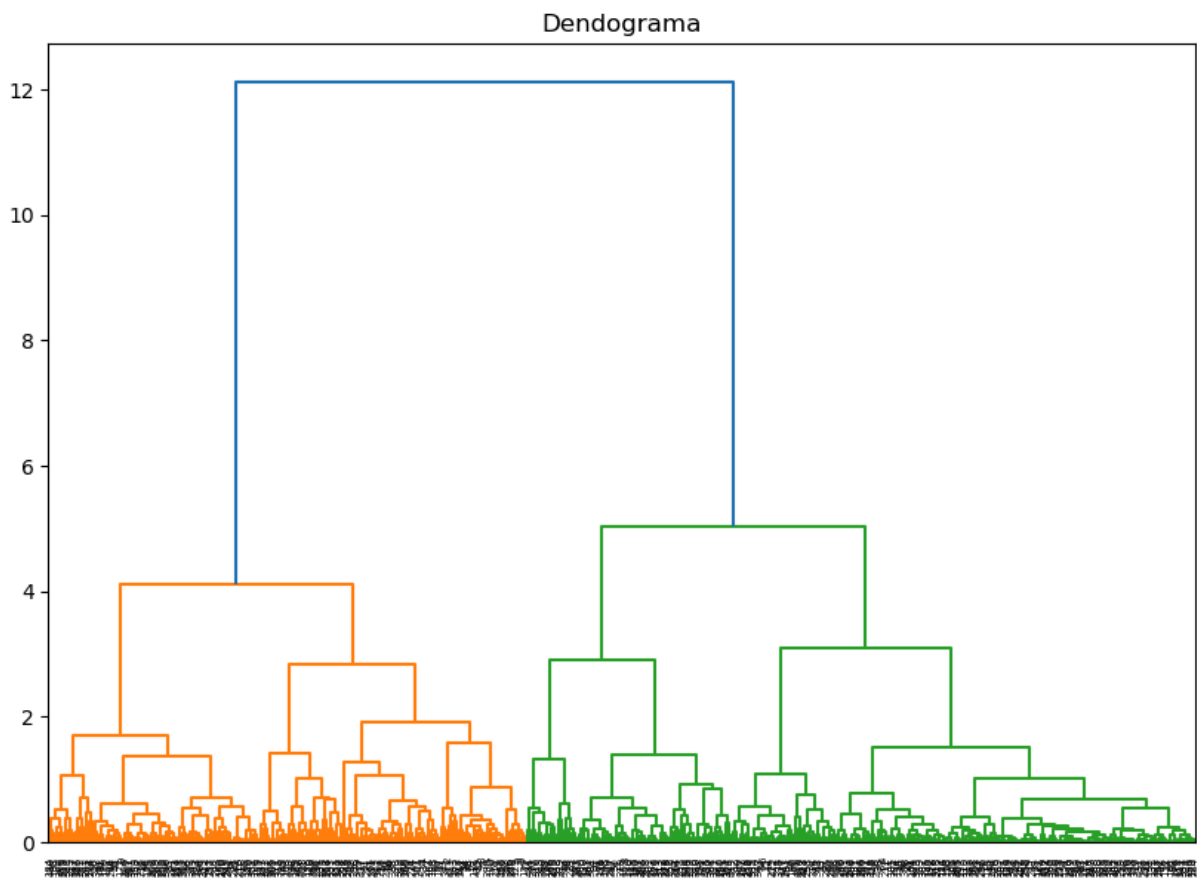
```
Out[7]: array([[1.11821406e-04, 1.67732109e-04, 7.08332695e-01, ...,
1.19648904e-02, 1.49505220e-01, 7.48085205e-02],
[1.25321880e-04, 1.87982820e-04, 4.42198253e-01, ...,
1.10408576e-01, 2.06342475e-01, 1.11285829e-01],
[1.24839188e-04, 1.87258782e-04, 3.96551681e-01, ...,
1.50119124e-01, 2.19467293e-01, 4.89619296e-01],
...,
[5.01633106e-05, 7.52449659e-05, 3.64461533e-01, ...,
1.09606834e-02, 3.72236846e-01, 4.68274505e-02],
[9.11309417e-05, 2.73392825e-04, 9.37737390e-01, ...,
9.45939175e-02, 1.53099982e-02, 1.93653251e-01],
[2.41225630e-04, 7.23676891e-04, 6.72295832e-01, ...,
1.56796660e-02, 1.15064626e-01, 1.25437328e-02]])
```

```
In [9]: data_scaled = pd.DataFrame(data_scaled, columns = data.columns)
data_scaled.head()
```

Out[9]:

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen
0	0.000112	0.000168	0.708333	0.539874	0.422741	0.011965	0.149505	0.074809
1	0.000125	0.000188	0.442198	0.614704	0.599540	0.110409	0.206342	0.111286
2	0.000125	0.000187	0.396552	0.549792	0.479632	0.150119	0.219467	0.489619
3	0.000065	0.000194	0.856837	0.077254	0.272650	0.413659	0.032749	0.115494
4	0.000079	0.000119	0.895416	0.214203	0.284997	0.155010	0.070358	0.205294

```
In [11]: import scipy.cluster.hierarchy as shc
plt.figure(figsize=(10,7))
plt.title("Dendograma")
dend = shc.dendrogram(shc.linkage(data_scaled, method='ward'))
```



```
In [13]: print(dend['color_list'])
```

[illegible]

```
In [15]: colores_unicos = set(dend['color_list'])
         colores_unicos
```

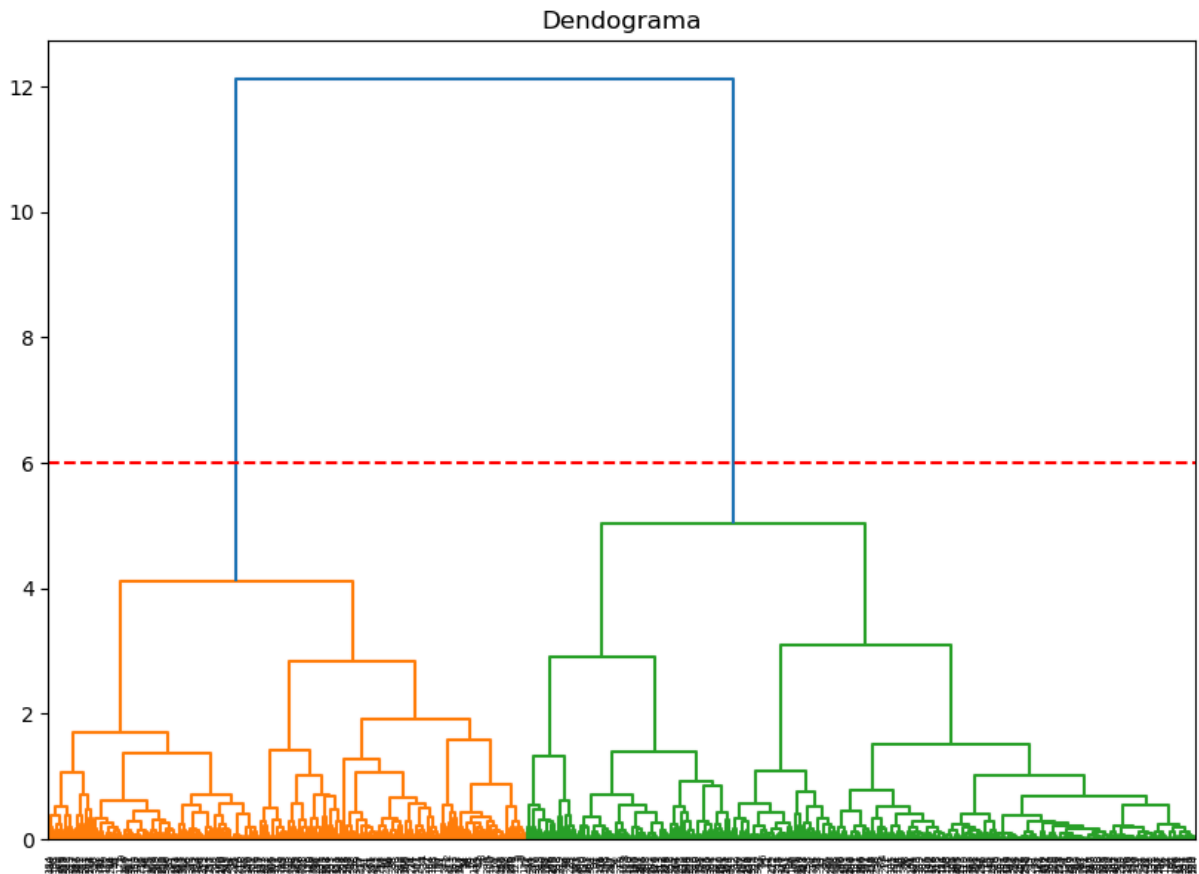
```
Out[15]: {'C0', 'C1', 'C2'}
```

```
In [23]: num_clusters_optimo = len(colores_unicos) - 1
          num_clusters_optimo
```

Out[23]: 2

```
In [25]: plt.figure(figsize=(10,7))
plt.title("Dendrograma")
dend = shc.dendrogram(shc.linkage(data_scaled, method='ward'))
plt.axhline(y=6, color = 'r', linestyle = '--')
```

```
Out[25]: <matplotlib.lines.Line2D at 0x1c918420d90>
```



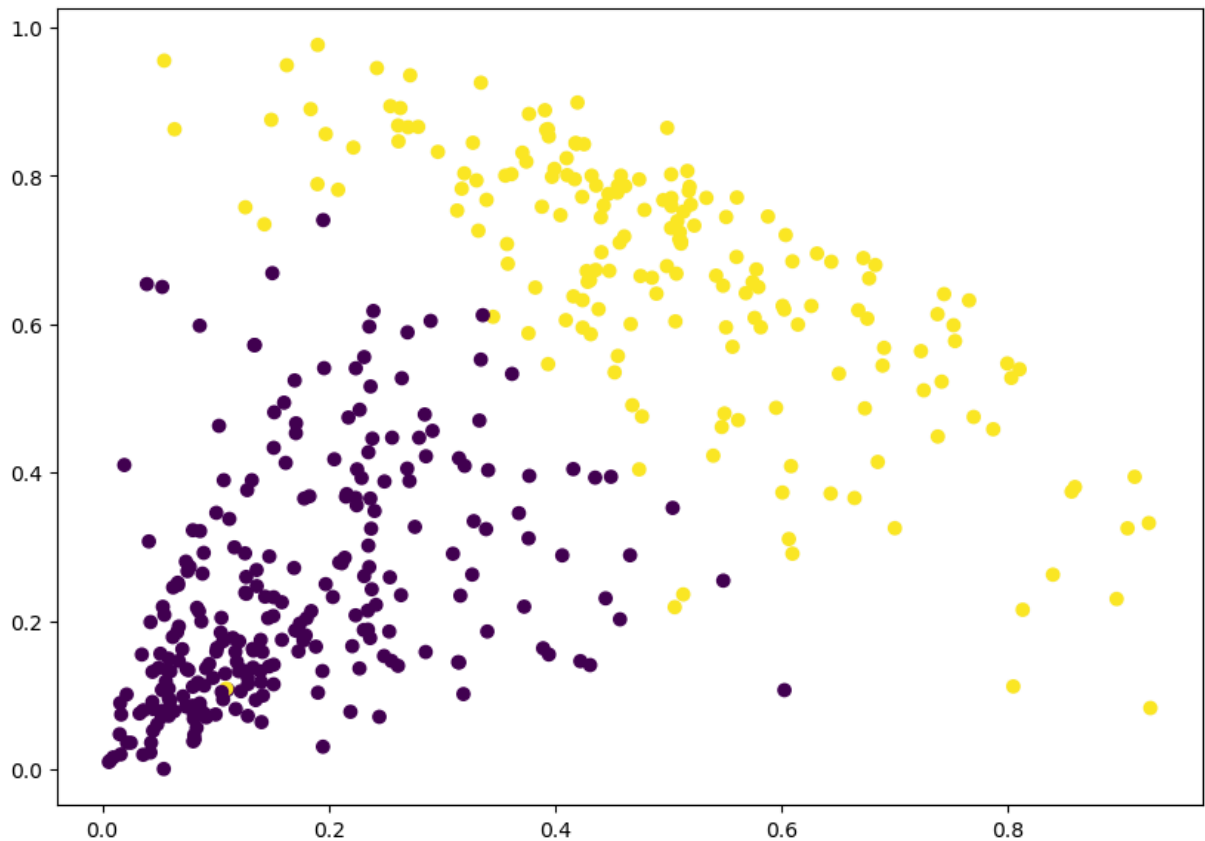
```
In [29]: from sklearn.cluster import AgglomerativeClustering
cluster = AgglomerativeClustering(n_clusters=2, metric='euclidean', linkage='ward')
grupos = cluster.fit_predict(data_scaled)
grupos
```

```
Out[29]: array([1, 1, 1, 0, 0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0,
                0, 1, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 0, 1, 1,
                1, 1, 1, 1, 1, 1, 0, 1, 0, 1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1,
                1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 1, 0, 1, 1, 0, 0, 1, 1, 0,
                0, 0, 0, 0, 1, 0, 1, 1, 1, 0, 0, 0, 1, 1, 1, 0, 0, 0, 1, 1, 1, 1,
                0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0,
                0, 0, 0, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1,
                0, 1, 1, 0, 1, 1, 1, 0, 0, 1, 1, 1, 1, 1, 0, 0, 1, 1, 1, 1, 1, 1,
                0, 0, 0, 1, 0, 0, 1, 1, 1, 0, 0, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1,
                0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0,
                0, 0, 1, 1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1,
                1, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0,
                0, 0, 0, 0, 1, 0, 1, 1, 0, 0, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0,
                0, 1, 0, 0, 1, 0, 0, 1, 0, 1, 0, 1, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0,
                0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 0, 1, 1, 0, 1, 1, 0, 1,
                1, 1, 0, 1, 0, 1, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0,
                0, 0, 1, 0, 1, 1, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 1, 1, 0, 0, 0,
                1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 1, 1, 1,
                1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 1],
                dtype=int64)
```

```
In [31]: plt.figure(figsize = (10,7))
```

```
plt.scatter(data_scaled['Milk'], data_scaled['Grocery'], c = cluster.labels_)
```

Out[31]: <matplotlib.collections.PathCollection at 0x1c917e7fd50>



```
In [33]: # Separación de grupos mediante PCA
campos = data_scaled.values
print(campos)
```

```
[[1.11821406e-04 1.67732109e-04 7.08332695e-01 ... 1.19648904e-02
 1.49505220e-01 7.48085205e-02]
 [1.25321880e-04 1.87982820e-04 4.42198253e-01 ... 1.10408576e-01
 2.06342475e-01 1.11285829e-01]
 [1.24839188e-04 1.87258782e-04 3.96551681e-01 ... 1.50119124e-01
 2.19467293e-01 4.89619296e-01]
 ...
 [5.01633106e-05 7.52449659e-05 3.64461533e-01 ... 1.09606834e-02
 3.72236846e-01 4.68274505e-02]
 [9.11309417e-05 2.73392825e-04 9.37737390e-01 ... 9.45939175e-02
 1.53099982e-02 1.93653251e-01]
 [2.41225630e-04 7.23676891e-04 6.72295832e-01 ... 1.56796660e-02
 1.15064626e-01 1.25437328e-02]]
```

```
In [35]: from sklearn import decomposition
pca = decomposition.PCA(n_components = 2)
pca.fit(campos)
campos = pca.transform(campos)
campos
```

```
Out[35]: array([[ 3.40370496e-02, -1.80420878e-01],
 [ 3.51057368e-01, -4.66441613e-02],
 [ 3.05375430e-01,  5.55510662e-02],
 [-3.83677396e-01,  1.61789121e-01],
 [-3.18591263e-01, -7.69700156e-02],
 [ 3.73478459e-02, -1.19851588e-01],
 [-1.20176243e-01, -2.43275779e-01],
 [ 2.39772413e-01, -1.04149634e-01],
 [ 1.65698845e-01, -1.87946102e-01],
 [ 5.68883753e-01, -1.09187885e-01],
 [ 5.50490278e-01,  8.66514506e-02],
 [-3.80404330e-01, -1.63301699e-01],
 [-2.16997115e-01, -2.19645753e-01],
 [-4.15793549e-02, -1.73397544e-01],
 [-1.33933983e-01, -2.36377213e-01],
 [-3.35298752e-01, -2.30915370e-01],
 [ 7.26283916e-01, -7.33583822e-02],
 [ 5.14896222e-02,  7.05034152e-03],
 [-1.39371504e-01, -1.51057486e-01],
 [ 1.64608998e-01, -2.12412480e-01],
 [-3.33835807e-01, -1.80358027e-01],
 [-3.23995148e-01,  2.39147842e-01],
 [-4.98705749e-01,  4.14682390e-02],
 [ 2.12232953e-01,  5.98882720e-03],
 [-6.30009465e-02, -1.35347961e-01],
 [-1.79665317e-01, -2.61548271e-01],
 [-4.14823639e-01,  3.42279073e-02],
 [-4.79249014e-01, -2.07953429e-01],
 [ 6.73129427e-01, -4.79248386e-02],
 [-5.70961913e-01, -1.91841970e-01],
 [-1.67202588e-01, -2.10793120e-01],
 [ 2.95611870e-01,  2.02389171e-01],
 [-5.17167853e-01, -2.21203305e-01],
 [-4.24196938e-01, -5.27010145e-02],
 [ 3.54771268e-01, -3.65412400e-02],
 [ 7.27927338e-01, -5.33108689e-02],
 [-4.52523392e-01, -1.65167176e-01],
 [ 1.29597534e-01, -1.53518969e-01],
 [ 6.18731741e-01, -9.98567672e-02],
 [-6.19228600e-01, -4.13006443e-02],
 [-4.13793874e-01,  1.28551576e-01],
 [-3.43063809e-01, -1.56418846e-01],
 [ 3.07519786e-01, -2.28921480e-01],
 [ 7.53223768e-01, -8.39338939e-02],
 [ 2.24695460e-01, -2.01776979e-01],
 [ 6.22063312e-01, -1.57586831e-02],
 [ 6.79846268e-01, -4.73480600e-02],
 [ 3.27795558e-01, -9.55443701e-02],
 [ 1.44410163e-01, -2.16552831e-01],
 [ 6.74077008e-01, -6.84096223e-02],
 [-3.15602398e-01,  2.61471315e-01],
 [ 4.43127098e-01, -1.51451615e-01],
 [-4.90754880e-01, -2.21463763e-01],
 [ 7.43017325e-01, -2.17414648e-02],
 [-5.70637048e-01, -1.32305756e-01],
 [ 1.61897456e-01, -1.02412575e-01],
```

[6.80069130e-01, -1.66615640e-02],
[5.03231325e-01, -1.48638052e-01],
[-5.01834682e-01, -1.78653901e-01],
[3.05604240e-01, -1.91464732e-01],
[1.00890901e-01, -2.67697600e-01],
[3.99561074e-01, -1.65831759e-01],
[1.40236863e-01, 8.28181774e-02],
[4.08870795e-01, -7.09228640e-02],
[-9.03889130e-02, 2.64962008e-01],
[7.69990379e-01, -1.09298565e-01],
[7.35890990e-01, -1.16366144e-01],
[-4.81786081e-02, -2.21148988e-01],
[3.25118853e-01, 4.99184692e-01],
[-3.82260364e-01, -1.29126685e-01],
[-4.34980462e-01, 3.00208993e-01],
[7.45489538e-02, -4.11025454e-02],
[4.17519668e-02, 6.42719663e-01],
[-2.62294212e-01, 8.45335699e-02],
[2.26029453e-01, -1.18599620e-01],
[-6.11228251e-01, -4.80484709e-04],
[4.29197018e-01, 6.15604204e-01],
[4.81706738e-01, -1.79409872e-01],
[-4.75041077e-01, -6.15378670e-02],
[4.76281160e-01, 4.24517554e-02],
[-2.98067142e-01, -2.11883345e-01],
[7.67546282e-01, -7.70688157e-02],
[2.30031687e-01, -1.54241186e-01],
[-5.50753086e-01, -1.17615669e-01],
[-1.94789047e-01, -1.90991741e-01],
[6.64313844e-01, -1.34824110e-01],
[4.54220988e-01, -2.80065613e-02],
[-4.38157051e-01, -7.06239740e-02],
[-2.33191454e-01, 5.23510639e-01],
[-4.38507986e-01, -1.00163161e-01],
[-5.27205290e-01, 3.63432139e-02],
[-3.84725272e-01, 3.24407654e-01],
[6.09426658e-01, -5.47808736e-02],
[-1.66045468e-01, 8.67796686e-01],
[5.02165348e-01, -1.12823857e-01],
[7.25007379e-01, 5.99899468e-03],
[7.55122640e-01, -1.08955508e-01],
[1.45940858e-01, 5.21804275e-01],
[1.06014532e-01, 4.87077633e-01],
[-4.01264649e-01, 2.60610397e-01],
[2.18121540e-01, -8.19292232e-02],
[7.16631603e-01, -2.25380234e-02],
[5.29503567e-01, 6.70270334e-02],
[-5.07251706e-01, 5.60710953e-02],
[-4.01188893e-01, -1.64911406e-01],
[-4.47945893e-01, -1.86856753e-01],
[6.92420411e-01, -1.12234792e-01],
[4.20871074e-01, -8.95874663e-02],
[6.35552091e-01, -2.48684459e-02],
[7.04382875e-01, -4.14433062e-02],
[-4.87851159e-01, -3.87639346e-02],
[3.22221124e-01, -1.79709017e-01],

[-4.73638478e-01, 1.90964168e-01],
[-4.89434312e-01, -1.29715397e-02],
[-5.45415376e-01, -8.96347007e-02],
[-4.40321505e-01, -3.02049954e-02],
[-3.59736185e-01, -1.20462857e-01],
[4.68667080e-03, -5.37226553e-02],
[-5.24106450e-01, 1.92511056e-02],
[-3.52313260e-01, 1.70316641e-01],
[-4.87069263e-01, -1.06634112e-01],
[-1.03450303e-01, 2.35158316e-01],
[-6.14165133e-01, -5.35381996e-02],
[1.29981880e-01, -6.24571595e-02],
[-5.67154960e-01, -9.65112949e-02],
[-5.62544329e-01, -2.43749455e-02],
[-5.28610348e-01, 1.60061911e-01],
[-1.69405331e-01, -2.07508734e-01],
[6.34852210e-01, 7.58407518e-02],
[-5.99064164e-01, -1.72225497e-01],
[-3.59331379e-01, 4.57567083e-01],
[-4.18374199e-01, -5.44280241e-02],
[-4.63713323e-01, -1.99604919e-01],
[-5.05456383e-01, -1.81808278e-01],
[-3.52285781e-01, -1.41640494e-01],
[-2.99690179e-01, -9.28211461e-02],
[3.63475981e-01, 2.16338091e-01],
[5.19735352e-01, -3.69874861e-02],
[-2.60365473e-01, -1.97639995e-01],
[5.97389341e-02, -6.20719662e-02],
[-2.70392295e-01, -1.30816527e-01],
[-3.38769193e-01, -1.73375038e-01],
[-4.25273239e-01, -1.57577342e-01],
[-4.34514025e-01, 4.15564262e-01],
[-4.82110726e-01, -2.06866073e-01],
[2.96333426e-01, -2.43962756e-01],
[-3.63983846e-01, -1.22613473e-01],
[-2.81369162e-01, -1.10147260e-01],
[-4.76397461e-01, 3.43703257e-01],
[-5.81542340e-01, -1.86786816e-01],
[-5.21303477e-01, -1.66920263e-01],
[3.94004028e-01, 9.95566822e-02],
[-5.06978600e-01, -9.04014501e-02],
[2.86859551e-01, 8.79092496e-02],
[-4.68027517e-01, -1.31938521e-01],
[7.12664171e-01, -1.12878705e-01],
[5.84135780e-01, -1.48282797e-01],
[-5.14701213e-01, -6.03229076e-02],
[5.86657814e-01, -6.50949747e-02],
[7.57866799e-01, -9.00865812e-02],
[6.49420759e-01, -8.90239974e-02],
[-5.90165441e-01, -1.08852374e-01],
[-4.30908934e-01, -8.27549896e-02],
[6.44506232e-01, -8.10786344e-02],
[4.01301513e-01, 1.04069927e-02],
[2.67136265e-01, -7.41762676e-02],
[4.53894565e-01, -8.54217066e-02],
[2.92116091e-01, -1.14922431e-01],

[-5.00801129e-01, -2.83747021e-03],
[-3.31542329e-01, 1.22155150e-01],
[7.59204668e-01, -3.36936733e-02],
[7.25036896e-01, 2.35040300e-02],
[6.88544230e-01, -5.11999893e-02],
[7.45287159e-01, -8.62137068e-02],
[5.56620944e-01, 3.12251305e-01],
[6.54122938e-01, -1.07705750e-01],
[-4.85880585e-01, -8.07793100e-02],
[-1.83084867e-01, -6.67240374e-02],
[-1.14072184e-01, -1.08654684e-01],
[1.42664711e-01, 4.75047756e-01],
[-7.02023479e-03, -1.95836482e-01],
[-4.20557019e-01, -8.00556892e-02],
[7.34681336e-01, -6.37764400e-02],
[5.57753564e-02, 3.86989624e-01],
[6.51175951e-01, -1.03707929e-01],
[-2.11688710e-01, 6.72622227e-02],
[-2.45868911e-01, 5.08336716e-01],
[5.31899526e-01, 2.93475419e-01],
[3.49267208e-01, -7.08877753e-02],
[7.18728186e-01, -2.26513597e-02],
[-4.03482871e-01, 1.70807142e-01],
[-5.65892127e-01, -1.65077097e-01],
[-2.72364466e-01, -1.56853473e-01],
[7.15311782e-01, -8.35088481e-02],
[-3.04046171e-01, -1.51663530e-01],
[-3.21755492e-01, 1.59089099e-01],
[-4.10688469e-01, 2.90331429e-01],
[6.26028727e-01, -4.55002502e-02],
[-2.04072537e-01, -2.44602809e-02],
[-3.96677960e-01, -1.79212302e-01],
[6.76762807e-01, 2.45055698e-03],
[6.53731361e-01, -3.78615644e-02],
[-2.14740140e-01, -2.92137031e-02],
[5.49864703e-01, -2.33037095e-02],
[3.60712191e-01, 5.57471920e-02],
[7.41904709e-01, -7.72892652e-02],
[-4.89609235e-01, -1.04097905e-01],
[4.95741946e-01, 1.07301179e-01],
[5.83529840e-01, 7.08027286e-02],
[5.87944401e-01, -1.19335783e-01],
[-5.43508714e-01, -1.16708607e-01],
[5.87376154e-01, -6.12470324e-02],
[-3.87794187e-01, -6.83325129e-02],
[4.98041281e-01, -2.38371319e-02],
[6.43969038e-01, -8.52640028e-02],
[6.58618380e-01, -4.16634553e-02],
[7.27041859e-01, -1.14339073e-01],
[-5.35775804e-01, -9.04180388e-02],
[7.30053541e-01, 1.10509462e-02],
[-3.77713716e-01, 2.07322220e-01],
[-5.56534312e-01, -1.07596282e-01],
[4.64168138e-01, -1.70026331e-01],
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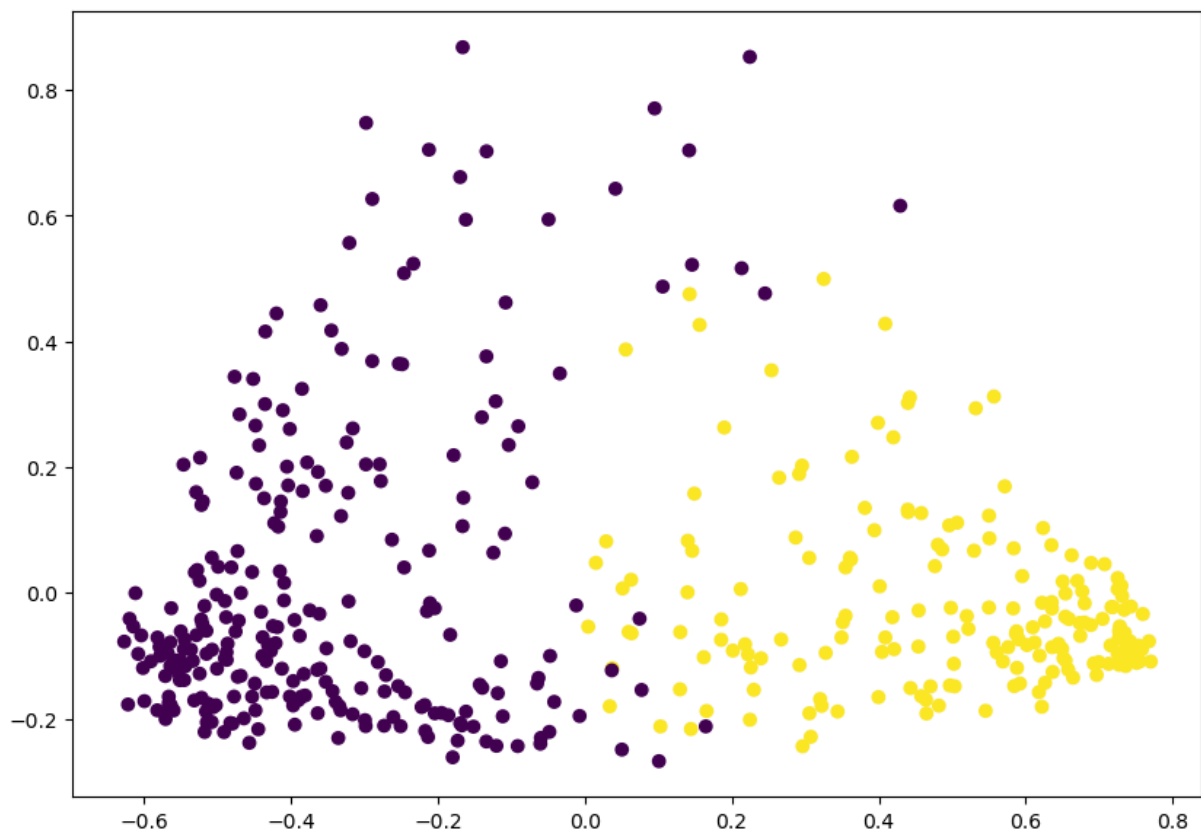
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```

```
In [37]: plt.figure(figsize=(10,7))
plt.scatter(campos[:,0], campos[:,1], c = cluster.labels_)
```

```
Out[37]: <matplotlib.collections.PathCollection at 0x1c9180ee150>
```



```
In [39]: dataframe = pd.DataFrame(grupos, columns=['grupo'])
dataframe
```

Out[39]:

grupo	
0	1
1	1
2	1
3	0
4	0
...	...
435	0
436	0
437	1
438	0
439	1

440 rows × 1 columns

```
In [41]: dataframe2 = pd.concat([data, dataframe], axis=1, join = 'inner')
dataframe2
```

Out[41]:

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	grup
0	2	3	12669	9656	7561	214	2674	1338	
1	2	3	7057	9810	9568	1762	3293	1776	
2	2	3	6353	8808	7684	2405	3516	7844	
3	1	3	13265	1196	4221	6404	507	1788	
4	2	3	22615	5410	7198	3915	1777	5185	
...	...	...	...	...	...	...	...	...	
435	1	3	29703	12051	16027	13135	182	2204	
436	1	3	39228	1431	764	4510	93	2346	
437	2	3	14531	15488	30243	437	14841	1867	
438	1	3	10290	1981	2232	1038	168	2125	
439	1	3	2787	1698	2510	65	477	52	

440 rows × 9 columns



In [43]:

```
pd.options.display.max_rows = None  
dataframe2
```

Out[43]:

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	gru
0	2	3	12669	9656	7561	214	2674	1338	
1	2	3	7057	9810	9568	1762	3293	1776	
2	2	3	6353	8808	7684	2405	3516	7844	
3	1	3	13265	1196	4221	6404	507	1788	
4	2	3	22615	5410	7198	3915	1777	5185	
5	2	3	9413	8259	5126	666	1795	1451	
6	2	3	12126	3199	6975	480	3140	545	
7	2	3	7579	4956	9426	1669	3321	2566	
8	1	3	5963	3648	6192	425	1716	750	
9	2	3	6006	11093	18881	1159	7425	2098	
10	2	3	3366	5403	12974	4400	5977	1744	
11	2	3	13146	1124	4523	1420	549	497	
12	2	3	31714	12319	11757	287	3881	2931	
13	2	3	21217	6208	14982	3095	6707	602	
14	2	3	24653	9465	12091	294	5058	2168	
15	1	3	10253	1114	3821	397	964	412	
16	2	3	1020	8816	12121	134	4508	1080	
17	1	3	5876	6157	2933	839	370	4478	
18	2	3	18601	6327	10099	2205	2767	3181	
19	1	3	7780	2495	9464	669	2518	501	
20	2	3	17546	4519	4602	1066	2259	2124	
21	1	3	5567	871	2010	3383	375	569	
22	1	3	31276	1917	4469	9408	2381	4334	
23	2	3	26373	36423	22019	5154	4337	16523	
24	2	3	22647	9776	13792	2915	4482	5778	
25	2	3	16165	4230	7595	201	4003	57	
26	1	3	9898	961	2861	3151	242	833	
27	1	3	14276	803	3045	485	100	518	
28	2	3	4113	20484	25957	1158	8604	5206	
29	1	3	43088	2100	2609	1200	1107	823	

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	gru
30	1	3	18815	3610	11107	1148	2134	2963	
31	1	3	2612	4339	3133	2088	820	985	
32	1	3	21632	1318	2886	266	918	405	
33	1	3	29729	4786	7326	6130	361	1083	
34	1	3	1502	1979	2262	425	483	395	
35	2	3	688	5491	11091	833	4239	436	
36	1	3	29955	4362	5428	1729	862	4626	
37	2	3	15168	10556	12477	1920	6506	714	
38	2	3	4591	15729	16709	33	6956	433	
39	1	3	56159	555	902	10002	212	2916	
40	1	3	24025	4332	4757	9510	1145	5864	
41	1	3	19176	3065	5956	2033	2575	2802	
42	2	3	10850	7555	14961	188	6899	46	
43	2	3	630	11095	23998	787	9529	72	
44	2	3	9670	7027	10471	541	4618	65	
45	2	3	5181	22044	21531	1740	7353	4985	
46	2	3	3103	14069	21955	1668	6792	1452	
47	2	3	44466	54259	55571	7782	24171	6465	
48	2	3	11519	6152	10868	584	5121	1476	
49	2	3	4967	21412	28921	1798	13583	1163	
50	1	3	6269	1095	1980	3860	609	2162	
51	1	3	3347	4051	6996	239	1538	301	
52	2	3	40721	3916	5876	532	2587	1278	
53	2	3	491	10473	11532	744	5611	224	
54	1	3	27329	1449	1947	2436	204	1333	
55	1	3	5264	3683	5005	1057	2024	1130	
56	2	3	4098	29892	26866	2616	17740	1340	
57	2	3	5417	9933	10487	38	7572	1282	
58	1	3	13779	1970	1648	596	227	436	
59	1	3	6137	5360	8040	129	3084	1603	

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	gru
60	2	3	8590	3045	7854	96	4095	225	
61	2	3	35942	38369	59598	3254	26701	2017	
62	2	3	7823	6245	6544	4154	4074	964	
63	2	3	9396	11601	15775	2896	7677	1295	
64	1	3	4760	1227	3250	3724	1247	1145	
65	2	3	85	20959	45828	36	24231	1423	
66	1	3	9	1534	7417	175	3468	27	
67	2	3	19913	6759	13462	1256	5141	834	
68	1	3	2446	7260	3993	5870	788	3095	
69	1	3	8352	2820	1293	779	656	144	
70	1	3	16705	2037	3202	10643	116	1365	
71	1	3	18291	1266	21042	5373	4173	14472	
72	1	3	4420	5139	2661	8872	1321	181	
73	2	3	19899	5332	8713	8132	764	648	
74	2	3	8190	6343	9794	1285	1901	1780	
75	1	3	20398	1137	3	4407	3	975	
76	1	3	717	3587	6532	7530	529	894	
77	2	3	12205	12697	28540	869	12034	1009	
78	1	3	10766	1175	2067	2096	301	167	
79	1	3	1640	3259	3655	868	1202	1653	
80	1	3	7005	829	3009	430	610	529	
81	2	3	219	9540	14403	283	7818	156	
82	2	3	10362	9232	11009	737	3537	2342	
83	1	3	20874	1563	1783	2320	550	772	
84	2	3	11867	3327	4814	1178	3837	120	
85	2	3	16117	46197	92780	1026	40827	2944	
86	2	3	22925	73498	32114	987	20070	903	
87	1	3	43265	5025	8117	6312	1579	14351	
88	1	3	7864	542	4042	9735	165	46	
89	1	3	24904	3836	5330	3443	454	3178	

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	gru
90	1	3	11405	596	1638	3347	69	360	
91	1	3	12754	2762	2530	8693	627	1117	
92	2	3	9198	27472	32034	3232	18906	5130	
93	1	3	11314	3090	2062	35009	71	2698	
94	2	3	5626	12220	11323	206	5038	244	
95	1	3	3	2920	6252	440	223	709	
96	2	3	23	2616	8118	145	3874	217	
97	1	3	403	254	610	774	54	63	
98	1	3	503	112	778	895	56	132	
99	1	3	9658	2182	1909	5639	215	323	
100	2	3	11594	7779	12144	3252	8035	3029	
101	2	3	1420	10810	16267	1593	6766	1838	
102	2	3	2932	6459	7677	2561	4573	1386	
103	1	3	56082	3504	8906	18028	1480	2498	
104	1	3	14100	2132	3445	1336	1491	548	
105	1	3	15587	1014	3970	910	139	1378	
106	2	3	1454	6337	10704	133	6830	1831	
107	2	3	8797	10646	14886	2471	8969	1438	
108	2	3	1531	8397	6981	247	2505	1236	
109	2	3	1406	16729	28986	673	836	3	
110	1	3	11818	1648	1694	2276	169	1647	
111	2	3	12579	11114	17569	805	6457	1519	
112	1	3	19046	2770	2469	8853	483	2708	
113	1	3	14438	2295	1733	3220	585	1561	
114	1	3	18044	1080	2000	2555	118	1266	
115	1	3	11134	793	2988	2715	276	610	
116	1	3	11173	2521	3355	1517	310	222	
117	1	3	6990	3880	5380	1647	319	1160	
118	1	3	20049	1891	2362	5343	411	933	
119	1	3	8258	2344	2147	3896	266	635	

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	gru
120	1	3	17160	1200	3412	2417	174	1136	
121	1	3	4020	3234	1498	2395	264	255	
122	1	3	12212	201	245	1991	25	860	
123	2	3	11170	10769	8814	2194	1976	143	
124	1	3	36050	1642	2961	4787	500	1621	
125	1	3	76237	3473	7102	16538	778	918	
126	1	3	19219	1840	1658	8195	349	483	
127	2	3	21465	7243	10685	880	2386	2749	
128	1	3	140	8847	3823	142	1062	3	
129	1	3	42312	926	1510	1718	410	1819	
130	1	3	7149	2428	699	6316	395	911	
131	1	3	2101	589	314	346	70	310	
132	1	3	14903	2032	2479	576	955	328	
133	1	3	9434	1042	1235	436	256	396	
134	1	3	7388	1882	2174	720	47	537	
135	1	3	6300	1289	2591	1170	199	326	
136	1	3	4625	8579	7030	4575	2447	1542	
137	1	3	3087	8080	8282	661	721	36	
138	1	3	13537	4257	5034	155	249	3271	
139	1	3	5387	4979	3343	825	637	929	
140	1	3	17623	4280	7305	2279	960	2616	
141	1	3	30379	13252	5189	321	51	1450	
142	1	3	37036	7152	8253	2995	20	3	
143	1	3	10405	1596	1096	8425	399	318	
144	1	3	18827	3677	1988	118	516	201	
145	2	3	22039	8384	34792	42	12591	4430	
146	1	3	7769	1936	2177	926	73	520	
147	1	3	9203	3373	2707	1286	1082	526	
148	1	3	5924	584	542	4052	283	434	
149	1	3	31812	1433	1651	800	113	1440	

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	gru
150	1	3	16225	1825	1765	853	170	1067	
151	1	3	1289	3328	2022	531	255	1774	
152	1	3	18840	1371	3135	3001	352	184	
153	1	3	3463	9250	2368	779	302	1627	
154	1	3	622	55	137	75	7	8	
155	2	3	1989	10690	19460	233	11577	2153	
156	2	3	3830	5291	14855	317	6694	3182	
157	1	3	17773	1366	2474	3378	811	418	
158	2	3	2861	6570	9618	930	4004	1682	
159	2	3	355	7704	14682	398	8077	303	
160	2	3	1725	3651	12822	824	4424	2157	
161	1	3	12434	540	283	1092	3	2233	
162	1	3	15177	2024	3810	2665	232	610	
163	2	3	5531	15726	26870	2367	13726	446	
164	2	3	5224	7603	8584	2540	3674	238	
165	2	3	15615	12653	19858	4425	7108	2379	
166	2	3	4822	6721	9170	993	4973	3637	
167	1	3	2926	3195	3268	405	1680	693	
168	1	3	5809	735	803	1393	79	429	
169	1	3	5414	717	2155	2399	69	750	
170	2	3	260	8675	13430	1116	7015	323	
171	2	3	200	25862	19816	651	8773	6250	
172	1	3	955	5479	6536	333	2840	707	
173	2	3	514	7677	19805	937	9836	716	
174	1	3	286	1208	5241	2515	153	1442	
175	2	3	2343	7845	11874	52	4196	1697	
176	1	3	45640	6958	6536	7368	1532	230	
177	1	3	12759	7330	4533	1752	20	2631	
178	1	3	11002	7075	4945	1152	120	395	
179	1	3	3157	4888	2500	4477	273	2165	

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	gru
180	1	3	12356	6036	8887	402	1382	2794	
181	1	3	112151	29627	18148	16745	4948	8550	
182	1	3	694	8533	10518	443	6907	156	
183	1	3	36847	43950	20170	36534	239	47943	
184	1	3	327	918	4710	74	334	11	
185	1	3	8170	6448	1139	2181	58	247	
186	1	3	3009	521	854	3470	949	727	
187	1	3	2438	8002	9819	6269	3459	3	
188	2	3	8040	7639	11687	2758	6839	404	
189	2	3	834	11577	11522	275	4027	1856	
190	1	3	16936	6250	1981	7332	118	64	
191	1	3	13624	295	1381	890	43	84	
192	1	3	5509	1461	2251	547	187	409	
193	2	3	180	3485	20292	959	5618	666	
194	1	3	7107	1012	2974	806	355	1142	
195	1	3	17023	5139	5230	7888	330	1755	
196	1	1	30624	7209	4897	18711	763	2876	
197	2	1	2427	7097	10391	1127	4314	1468	
198	1	1	11686	2154	6824	3527	592	697	
199	1	1	9670	2280	2112	520	402	347	
200	2	1	3067	13240	23127	3941	9959	731	
201	2	1	4484	14399	24708	3549	14235	1681	
202	1	1	25203	11487	9490	5065	284	6854	
203	1	1	583	685	2216	469	954	18	
204	1	1	1956	891	5226	1383	5	1328	
205	2	1	1107	11711	23596	955	9265	710	
206	1	1	6373	780	950	878	288	285	
207	2	1	2541	4737	6089	2946	5316	120	
208	1	1	1537	3748	5838	1859	3381	806	
209	2	1	5550	12729	16767	864	12420	797	

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	gru
210	1	1	18567	1895	1393	1801	244	2100	
211	2	1	12119	28326	39694	4736	19410	2870	
212	1	1	7291	1012	2062	1291	240	1775	
213	1	1	3317	6602	6861	1329	3961	1215	
214	2	1	2362	6551	11364	913	5957	791	
215	1	1	2806	10765	15538	1374	5828	2388	
216	2	1	2532	16599	36486	179	13308	674	
217	1	1	18044	1475	2046	2532	130	1158	
218	2	1	18	7504	15205	1285	4797	6372	
219	1	1	4155	367	1390	2306	86	130	
220	1	1	14755	899	1382	1765	56	749	
221	1	1	5396	7503	10646	91	4167	239	
222	1	1	5041	1115	2856	7496	256	375	
223	2	1	2790	2527	5265	5612	788	1360	
224	1	1	7274	659	1499	784	70	659	
225	1	1	12680	3243	4157	660	761	786	
226	2	1	20782	5921	9212	1759	2568	1553	
227	1	1	4042	2204	1563	2286	263	689	
228	1	1	1869	577	572	950	4762	203	
229	1	1	8656	2746	2501	6845	694	980	
230	2	1	11072	5989	5615	8321	955	2137	
231	1	1	2344	10678	3828	1439	1566	490	
232	1	1	25962	1780	3838	638	284	834	
233	1	1	964	4984	3316	937	409	7	
234	1	1	15603	2703	3833	4260	325	2563	
235	1	1	1838	6380	2824	1218	1216	295	
236	1	1	8635	820	3047	2312	415	225	
237	1	1	18692	3838	593	4634	28	1215	
238	1	1	7363	475	585	1112	72	216	
239	1	1	47493	2567	3779	5243	828	2253	

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	gru
240	1	1	22096	3575	7041	11422	343	2564	
241	1	1	24929	1801	2475	2216	412	1047	
242	1	1	18226	659	2914	3752	586	578	
243	1	1	11210	3576	5119	561	1682	2398	
244	1	1	6202	7775	10817	1183	3143	1970	
245	2	1	3062	6154	13916	230	8933	2784	
246	1	1	8885	2428	1777	1777	430	610	
247	1	1	13569	346	489	2077	44	659	
248	1	1	15671	5279	2406	559	562	572	
249	1	1	8040	3795	2070	6340	918	291	
250	1	1	3191	1993	1799	1730	234	710	
251	2	1	6134	23133	33586	6746	18594	5121	
252	1	1	6623	1860	4740	7683	205	1693	
253	1	1	29526	7961	16966	432	363	1391	
254	1	1	10379	17972	4748	4686	1547	3265	
255	1	1	31614	489	1495	3242	111	615	
256	1	1	11092	5008	5249	453	392	373	
257	1	1	8475	1931	1883	5004	3593	987	
258	1	1	56083	4563	2124	6422	730	3321	
259	1	1	53205	4959	7336	3012	967	818	
260	1	1	9193	4885	2157	327	780	548	
261	1	1	7858	1110	1094	6818	49	287	
262	1	1	23257	1372	1677	982	429	655	
263	1	1	2153	1115	6684	4324	2894	411	
264	2	1	1073	9679	15445	61	5980	1265	
265	1	1	5909	23527	13699	10155	830	3636	
266	2	1	572	9763	22182	2221	4882	2563	
267	1	1	20893	1222	2576	3975	737	3628	
268	2	1	11908	8053	19847	1069	6374	698	
269	1	1	15218	258	1138	2516	333	204	

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	gru
270	1	1	4720	1032	975	5500	197	56	
271	1	1	2083	5007	1563	1120	147	1550	
272	1	1	514	8323	6869	529	93	1040	
273	1	3	36817	3045	1493	4802	210	1824	
274	1	3	894	1703	1841	744	759	1153	
275	1	3	680	1610	223	862	96	379	
276	1	3	27901	3749	6964	4479	603	2503	
277	1	3	9061	829	683	16919	621	139	
278	1	3	11693	2317	2543	5845	274	1409	
279	2	3	17360	6200	9694	1293	3620	1721	
280	1	3	3366	2884	2431	977	167	1104	
281	2	3	12238	7108	6235	1093	2328	2079	
282	1	3	49063	3965	4252	5970	1041	1404	
283	1	3	25767	3613	2013	10303	314	1384	
284	1	3	68951	4411	12609	8692	751	2406	
285	1	3	40254	640	3600	1042	436	18	
286	1	3	7149	2247	1242	1619	1226	128	
287	1	3	15354	2102	2828	8366	386	1027	
288	1	3	16260	594	1296	848	445	258	
289	1	3	42786	286	471	1388	32	22	
290	1	3	2708	2160	2642	502	965	1522	
291	1	3	6022	3354	3261	2507	212	686	
292	1	3	2838	3086	4329	3838	825	1060	
293	2	2	3996	11103	12469	902	5952	741	
294	1	2	21273	2013	6550	909	811	1854	
295	2	2	7588	1897	5234	417	2208	254	
296	1	2	19087	1304	3643	3045	710	898	
297	2	2	8090	3199	6986	1455	3712	531	
298	2	2	6758	4560	9965	934	4538	1037	
299	1	2	444	879	2060	264	290	259	

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	gru
300	2	2	16448	6243	6360	824	2662	2005	
301	2	2	5283	13316	20399	1809	8752	172	
302	2	2	2886	5302	9785	364	6236	555	
303	2	2	2599	3688	13829	492	10069	59	
304	2	2	161	7460	24773	617	11783	2410	
305	2	2	243	12939	8852	799	3909	211	
306	2	2	6468	12867	21570	1840	7558	1543	
307	1	2	17327	2374	2842	1149	351	925	
308	1	2	6987	1020	3007	416	257	656	
309	2	2	918	20655	13567	1465	6846	806	
310	1	2	7034	1492	2405	12569	299	1117	
311	1	2	29635	2335	8280	3046	371	117	
312	2	2	2137	3737	19172	1274	17120	142	
313	1	2	9784	925	2405	4447	183	297	
314	1	2	10617	1795	7647	1483	857	1233	
315	2	2	1479	14982	11924	662	3891	3508	
316	1	2	7127	1375	2201	2679	83	1059	
317	1	2	1182	3088	6114	978	821	1637	
318	1	2	11800	2713	3558	2121	706	51	
319	2	2	9759	25071	17645	1128	12408	1625	
320	1	2	1774	3696	2280	514	275	834	
321	1	2	9155	1897	5167	2714	228	1113	
322	1	2	15881	713	3315	3703	1470	229	
323	1	2	13360	944	11593	915	1679	573	
324	1	2	25977	3587	2464	2369	140	1092	
325	1	2	32717	16784	13626	60869	1272	5609	
326	1	2	4414	1610	1431	3498	387	834	
327	1	2	542	899	1664	414	88	522	
328	1	2	16933	2209	3389	7849	210	1534	
329	1	2	5113	1486	4583	5127	492	739	

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	gru
330	1	2	9790	1786	5109	3570	182	1043	
331	2	2	11223	14881	26839	1234	9606	1102	
332	1	2	22321	3216	1447	2208	178	2602	
333	2	2	8565	4980	67298	131	38102	1215	
334	2	2	16823	928	2743	11559	332	3486	
335	2	2	27082	6817	10790	1365	4111	2139	
336	1	2	13970	1511	1330	650	146	778	
337	1	2	9351	1347	2611	8170	442	868	
338	1	2	3	333	7021	15601	15	550	
339	1	2	2617	1188	5332	9584	573	1942	
340	2	3	381	4025	9670	388	7271	1371	
341	2	3	2320	5763	11238	767	5162	2158	
342	1	3	255	5758	5923	349	4595	1328	
343	2	3	1689	6964	26316	1456	15469	37	
344	1	3	3043	1172	1763	2234	217	379	
345	1	3	1198	2602	8335	402	3843	303	
346	2	3	2771	6939	15541	2693	6600	1115	
347	2	3	27380	7184	12311	2809	4621	1022	
348	1	3	3428	2380	2028	1341	1184	665	
349	2	3	5981	14641	20521	2005	12218	445	
350	1	3	3521	1099	1997	1796	173	995	
351	2	3	1210	10044	22294	1741	12638	3137	
352	1	3	608	1106	1533	830	90	195	
353	2	3	117	6264	21203	228	8682	1111	
354	1	3	14039	7393	2548	6386	1333	2341	
355	1	3	190	727	2012	245	184	127	
356	1	3	22686	134	218	3157	9	548	
357	2	3	37	1275	22272	137	6747	110	
358	1	3	759	18664	1660	6114	536	4100	
359	1	3	796	5878	2109	340	232	776	

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	gru
360	1	3	19746	2872	2006	2601	468	503	
361	1	3	4734	607	864	1206	159	405	
362	1	3	2121	1601	2453	560	179	712	
363	1	3	4627	997	4438	191	1335	314	
364	1	3	2615	873	1524	1103	514	468	
365	2	3	4692	6128	8025	1619	4515	3105	
366	1	3	9561	2217	1664	1173	222	447	
367	1	3	3477	894	534	1457	252	342	
368	1	3	22335	1196	2406	2046	101	558	
369	1	3	6211	337	683	1089	41	296	
370	2	3	39679	3944	4955	1364	523	2235	
371	1	3	20105	1887	1939	8164	716	790	
372	1	3	3884	3801	1641	876	397	4829	
373	2	3	15076	6257	7398	1504	1916	3113	
374	1	3	6338	2256	1668	1492	311	686	
375	1	3	5841	1450	1162	597	476	70	
376	2	3	3136	8630	13586	5641	4666	1426	
377	1	3	38793	3154	2648	1034	96	1242	
378	1	3	3225	3294	1902	282	68	1114	
379	2	3	4048	5164	10391	130	813	179	
380	1	3	28257	944	2146	3881	600	270	
381	1	3	17770	4591	1617	9927	246	532	
382	1	3	34454	7435	8469	2540	1711	2893	
383	1	3	1821	1364	3450	4006	397	361	
384	1	3	10683	21858	15400	3635	282	5120	
385	1	3	11635	922	1614	2583	192	1068	
386	1	3	1206	3620	2857	1945	353	967	
387	1	3	20918	1916	1573	1960	231	961	
388	1	3	9785	848	1172	1677	200	406	
389	1	3	9385	1530	1422	3019	227	684	

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	gru
390	1	3	3352	1181	1328	5502	311	1000	
391	1	3	2647	2761	2313	907	95	1827	
392	1	3	518	4180	3600	659	122	654	
393	1	3	23632	6730	3842	8620	385	819	
394	1	3	12377	865	3204	1398	149	452	
395	1	3	9602	1316	1263	2921	841	290	
396	2	3	4515	11991	9345	2644	3378	2213	
397	1	3	11535	1666	1428	6838	64	743	
398	1	3	11442	1032	582	5390	74	247	
399	1	3	9612	577	935	1601	469	375	
400	1	3	4446	906	1238	3576	153	1014	
401	1	3	27167	2801	2128	13223	92	1902	
402	1	3	26539	4753	5091	220	10	340	
403	1	3	25606	11006	4604	127	632	288	
404	1	3	18073	4613	3444	4324	914	715	
405	1	3	6884	1046	1167	2069	593	378	
406	1	3	25066	5010	5026	9806	1092	960	
407	2	3	7362	12844	18683	2854	7883	553	
408	2	3	8257	3880	6407	1646	2730	344	
409	1	3	8708	3634	6100	2349	2123	5137	
410	1	3	6633	2096	4563	1389	1860	1892	
411	1	3	2126	3289	3281	1535	235	4365	
412	1	3	97	3605	12400	98	2970	62	
413	1	3	4983	4859	6633	17866	912	2435	
414	1	3	5969	1990	3417	5679	1135	290	
415	2	3	7842	6046	8552	1691	3540	1874	
416	2	3	4389	10940	10908	848	6728	993	
417	1	3	5065	5499	11055	364	3485	1063	
418	2	3	660	8494	18622	133	6740	776	
419	1	3	8861	3783	2223	633	1580	1521	

	Channel	Region	Fresh	Milk	Grocery	Frozen	Detergents_Paper	Delicassen	gru
420	1	3	4456	5266	13227	25	6818	1393	
421	2	3	17063	4847	9053	1031	3415	1784	
422	1	3	26400	1377	4172	830	948	1218	
423	2	3	17565	3686	4657	1059	1803	668	
424	2	3	16980	2884	12232	874	3213	249	
425	1	3	11243	2408	2593	15348	108	1886	
426	1	3	13134	9347	14316	3141	5079	1894	
427	1	3	31012	16687	5429	15082	439	1163	
428	1	3	3047	5970	4910	2198	850	317	
429	1	3	8607	1750	3580	47	84	2501	
430	1	3	3097	4230	16483	575	241	2080	
431	1	3	8533	5506	5160	13486	1377	1498	
432	1	3	21117	1162	4754	269	1328	395	
433	1	3	1982	3218	1493	1541	356	1449	
434	1	3	16731	3922	7994	688	2371	838	
435	1	3	29703	12051	16027	13135	182	2204	
436	1	3	39228	1431	764	4510	93	2346	
437	2	3	14531	15488	30243	437	14841	1867	
438	1	3	10290	1981	2232	1038	168	2125	
439	1	3	2787	1698	2510	65	477	52	

ANALISIS DEL CASO DE ESTUDIANTES

```
In [74]: data = pd.DataFrame(np.array([[7, 6.5, 9.2, 8.6, 8],
                                     [7.5, 9.4, 7.3, 7, 7],
                                     [7.6, 9.2, 8, 8, 7.5],
                                     [5, 6.5, 6.5, 7, 9],
                                     [6, 6, 7.8, 8.9, 7.3],
                                     [7.8, 9.6, 7.7, 8, 6.5],
                                     [6.3, 6.4, 8.2, 9, 7.2],
                                     [7.9, 9.7, 7.5, 8, 6],
                                     [6, 6, 6.5, 5.5, 8.7],
                                     [6.8, 7.2, 8.7, 9, 7]]),
                             #index=['Lucia', 'Pedro', 'Ines', 'Luis', 'Andres', 'Ana', 'Carlo
```

```
columns=['Matematicas', 'Ciencias', 'Español', 'Historia', 'EdFis']  
data
```

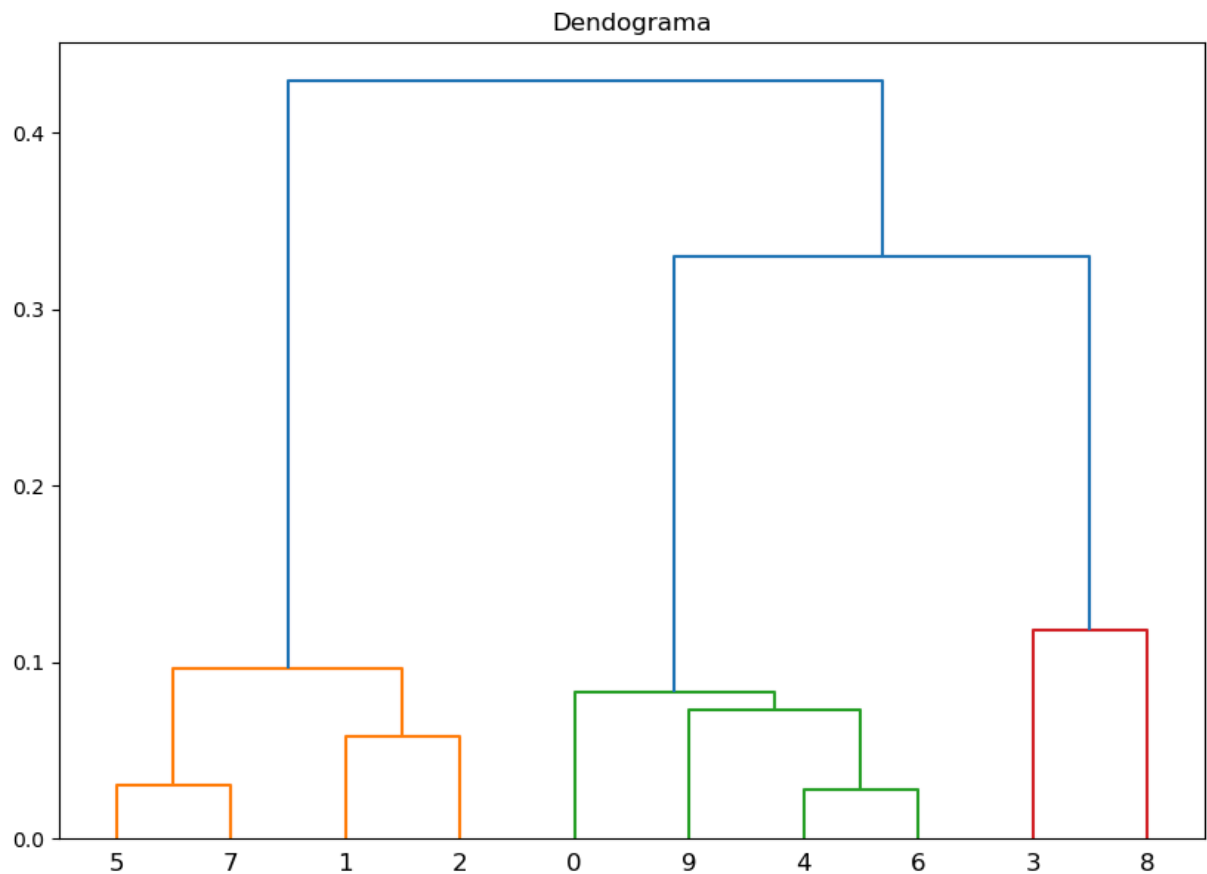
```
Out[74]:
```

	Matematicas	Ciencias	Español	Historia	EdFisica
0	7.0	6.5	9.2	8.6	8.0
1	7.5	9.4	7.3	7.0	7.0
2	7.6	9.2	8.0	8.0	7.5
3	5.0	6.5	6.5	7.0	9.0
4	6.0	6.0	7.8	8.9	7.3
5	7.8	9.6	7.7	8.0	6.5
6	6.3	6.4	8.2	9.0	7.2
7	7.9	9.7	7.5	8.0	6.0
8	6.0	6.0	6.5	5.5	8.7
9	6.8	7.2	8.7	9.0	7.0

```
In [76]: from sklearn.preprocessing import normalize  
data_scaled = normalize(data)  
data_scaled
```

```
Out[76]: array([[0.39512724, 0.36690387, 0.51931009, 0.48544204, 0.45157399],  
                [0.4360023 , 0.54645622, 0.42437557, 0.40693548, 0.40693548],  
                [0.42050606, 0.50903365, 0.44263796, 0.44263796, 0.41497309],  
                [0.32308533, 0.42001094, 0.42001094, 0.45231947, 0.5815536 ],  
                [0.36834085, 0.36834085, 0.47884311, 0.54637226, 0.44814804],  
                [0.43703137, 0.53788477, 0.43142841, 0.44823731, 0.36419281],  
                [0.37600718, 0.38197555, 0.48940617, 0.53715311, 0.42972249],  
                [0.44671284, 0.54849551, 0.42409446, 0.45236743, 0.33927557],  
                [0.40434535, 0.40434535, 0.4380408 , 0.37064991, 0.58630076],  
                [0.39015437, 0.41310462, 0.49916809, 0.51638078, 0.4016295 ]])
```

```
In [78]: import scipy.cluster.hierarchy as shc  
plt.figure(figsize=(10,7))  
plt.title("Dendograma")  
dend = shc.dendrogram(shc.linkage(data_scaled, method='ward'))
```

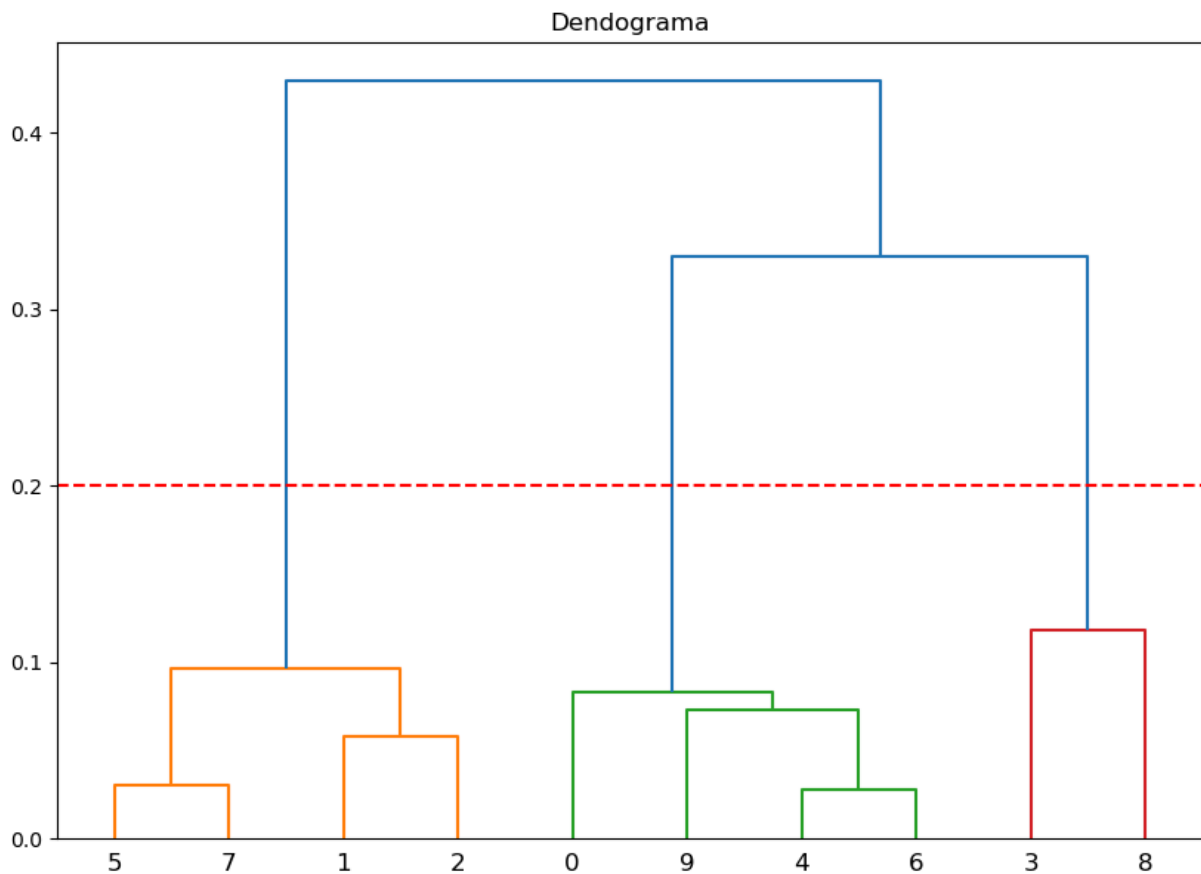


```
In [80]: colores_unicos = set(dend['color_list'])
num_clusters_optimo = len(colores_unicos)-1
num_clusters_optimo
```

Out[80]: 3

```
In [82]: plt.figure(figsize=(10,7))
plt.title("Dendrograma")
dend = shc.dendrogram(shc.linkage(data_scaled, method = 'ward'))
plt.axhline(y=0.2, color = 'r', linestyle = '--')
```

Out[82]: <matplotlib.lines.Line2D at 0x1c919e73d10>



```
In [84]: # Análisis gráfico con PCA
campos = data_scaled
from sklearn import decomposition
pca = decomposition.PCA(n_components = 2)
pca.fit(campos)
campos = pca.transform(campos)
campos
```

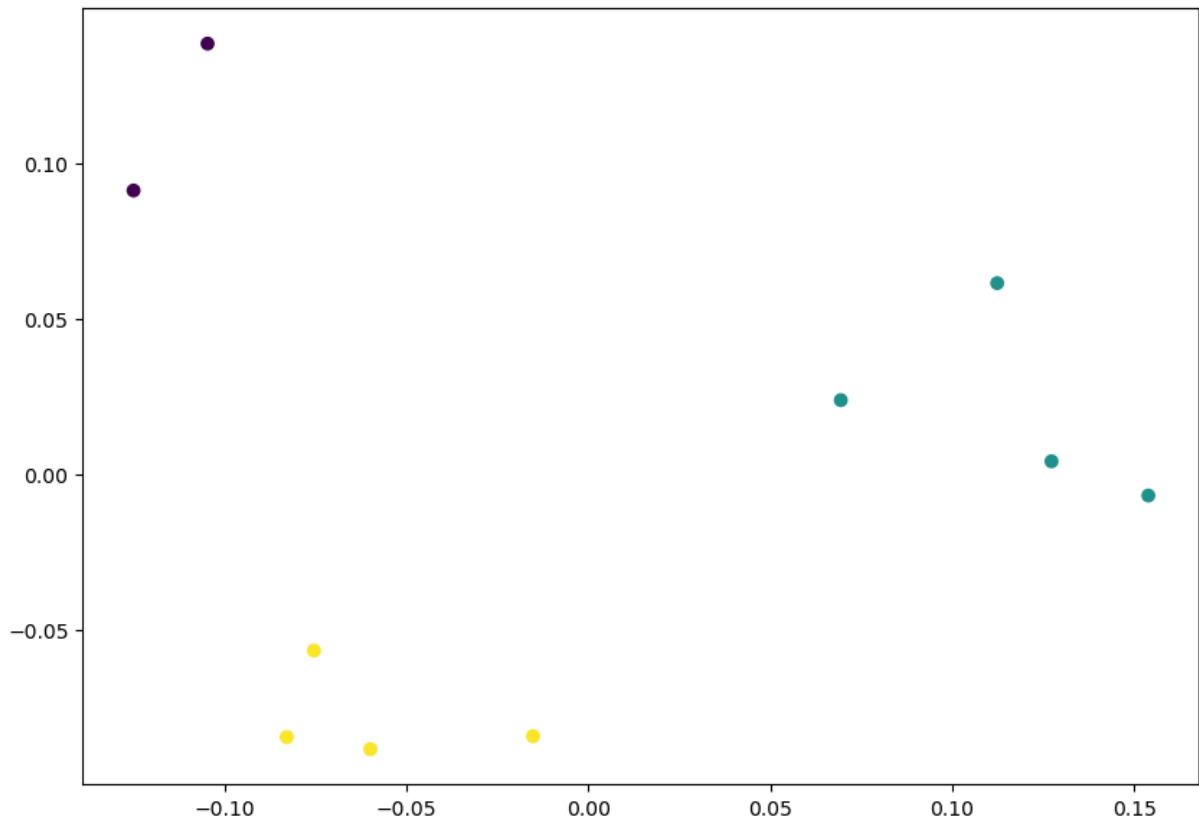
```
Out[84]: array([[ -0.07537384, -0.05649076],
 [ 0.11233306,  0.06156361],
 [ 0.06937082,  0.02395785],
 [-0.12493186,  0.09133353],
 [-0.08285196, -0.08431667],
 [ 0.12725973,  0.00431553],
 [-0.05987387, -0.08818057],
 [ 0.15385473, -0.00668172],
 [-0.1045932 ,  0.13852098],
 [-0.01519363, -0.08402179]])
```

```
In [86]: from sklearn.cluster import AgglomerativeClustering
cluster = AgglomerativeClustering(n_clusters=3, metric='euclidean', linkage='ward')
grupos = cluster.fit_predict(data_scaled)
grupos
```

```
Out[86]: array([2, 1, 1, 0, 2, 1, 2, 1, 0, 2], dtype=int64)
```

```
In [88]: plt.figure(figsize=(10,7))
plt.scatter(campos[:,0], campos[:,1], c = cluster.labels_)
```

Out[88]: <matplotlib.collections.PathCollection at 0x1c919e66150>



```
In [90]: dataframe = pd.DataFrame(grupos, columns=['grupo'])  
dataframe
```

Out[90]:

grupo	
0	2
1	1
2	1
3	0
4	2
5	1
6	2
7	1
8	0
9	2

```
In [92]: dataframe2 = pd.concat([data, dataframe], axis=1, join = 'inner')  
dataframe2
```

Out[92]:

	Matematicas	Ciencias	Español	Historia	EdFisica	grupo
0	7.0	6.5	9.2	8.6	8.0	2
1	7.5	9.4	7.3	7.0	7.0	1
2	7.6	9.2	8.0	8.0	7.5	1
3	5.0	6.5	6.5	7.0	9.0	0
4	6.0	6.0	7.8	8.9	7.3	2
5	7.8	9.6	7.7	8.0	6.5	1
6	6.3	6.4	8.2	9.0	7.2	2
7	7.9	9.7	7.5	8.0	6.0	1
8	6.0	6.0	6.5	5.5	8.7	0
9	6.8	7.2	8.7	9.0	7.0	2

```
In [94]: dataframe2.index = ['Lucia', 'Pedro', 'Ines', 'Luis', 'Andres', 'Ana', 'Carlos', 'J  
dataframe2
```

Out[94]:

	Matematicas	Ciencias	Español	Historia	EdFisica	grupo
Lucia	7.0	6.5	9.2	8.6	8.0	2
Pedro	7.5	9.4	7.3	7.0	7.0	1
Ines	7.6	9.2	8.0	8.0	7.5	1
Luis	5.0	6.5	6.5	7.0	9.0	0
Andres	6.0	6.0	7.8	8.9	7.3	2
Ana	7.8	9.6	7.7	8.0	6.5	1
Carlos	6.3	6.4	8.2	9.0	7.2	2
Jose	7.9	9.7	7.5	8.0	6.0	1
Sonia	6.0	6.0	6.5	5.5	8.7	0
Maria	6.8	7.2	8.7	9.0	7.0	2